" OrbitMind AI– Intelligent Productivity & Study Companion"

by

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Development Roadmap

Course Title: Software Development III: AI-Based Software Project Development

Course Code: CSE 4104

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Submitted to

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1.1 Team Information Sheet

Field	Details
Team Name	CSE4104-7D-T04
Section	7D
Project Title	OrbitMind AI — Smart Productivity & Study Assistant
Team Leader	Md. Imroz Newaz (ID: 11220321052)

1.1.1 Team Members

Member Name	Student ID	Role
Md. Imroz Newaz	11220321052	Team Leader & Database Manager
Zeniya Naznin	11230121138	Frontend Developer
Fatema Parvin Kanta	11230121192	Backend Developer
Md. Mahfuzur Rahman Hridoy	11230121186	AI Integration Lead

Project Information

Overview: OrbitMind AI is a comprehensive productivity web application designed to alleviate the academic pressures faced by university students. The system seamlessly integrates standard task management—such as deadline tracking and workload visualization—with advanced Artificial Intelligence powered by Google Gemini. Our platform acts as a proactive academic assistant: it automates the creation of day-by-day study roadmaps, summarizes dense lecture transcripts into easily digestible concepts, and features a conversational AI coach to guide students through complex topics. By finalizing our UI/UX designs, we are ensuring that this powerful functionality is delivered through an accessible, highly intuitive, and distraction-free interface, which is critical for maintaining student focus and reducing cognitive overload.

Development Roadmap

To ensure a smooth transition from the design phase into full system implementation, our team has established the following agile development roadmap for the remainder of the semester.

- **Week 06: Core Backend & Database Initialization**
    - Set up the Node.js/Express.js server architecture.
    - Initialize the MongoDB Atlas database and define Mongoose schemas (User, Task, Study Plan).
    - Develop the JWT-based authentication APIs (Register/Login).

- **Week 07: Core Frontend & State Management Development**
    - Connect the existing Vanilla HTML/CSS/JS frontend to the backend APIs.
    - Implement client-side session management and route protection.
    - Finalize the dynamic rendering of the user Dashboard and Task Board.

- **Week 08: AI Services Integration**
    - Securely connect the backend to the Google Gemini API.
    - Develop the prompt engineering logic for the AI Study Planner and AI Summarizer.
    - Connect the frontend Chatbot UI to the backend AI streaming routes.

- **Week 09: Advanced Feature Completion**
    - Implement the Notification system and workload analytics charts.
    - Finalize data validation and error handling across all forms.
    - Ensure the UI is fully responsive across mobile and desktop breakpoints.

- **Week 10: System Testing and Debugging**

- Conduct rigorous unit testing on backend API routes using Postman.
- Perform End-to-End (E2E) UI testing to ensure user flows operate flawlessly.
- Resolve any integration bugs between the database and the AI module.

- **Week 11: Final Deployment & Presentation Prep**
  - Deploy the Node.js backend and MongoDB database to a cloud provider (e.g., Render).
  - Host the static frontend interface.
  - Conduct final user acceptance testing and prepare the project defense presentation.